

School: SET		Batch: 2024-2028																					
Program: B Tech		Current Academic Year: 2026-27																					
Branch:		Semester: V																					
1	Course Code	CSE356																					
2	Course Title	Design and Analysis of Algorithms																					
3	Credits	3																					
4	Contact Hours (L-T-P-SL)	3-0-0-3																					
	Course Status	Core																					
5	Course Objective	Objective of this course is to 1. Reinforce basic design concepts (e.g., pseudocode, specifications, top-down design) 2. Knowledge of algorithm design strategies 3. Familiarity with an assortment of important algorithms. 4. Enable students to analyze time and space complexity																					
6	Course Outcomes	Students will be able to: CO1: Analyze the asymptotic performance of algorithms CO2: Describe the dynamic-programming and Greedy paradigm and explain when an algorithmic design situation calls for it. CO3: Demonstrate a familiarity with major algorithms and data structures CO4: Apply important algorithmic design paradigms and methods of analysis CO5: Discuss NP-complete problems and develop algorithms to solve the problems. CO6: Choose appropriate algorithm design techniques for solving problems.																					
7	Course Description	This course introduces concepts related to the design and analysis of algorithms. Specifically, it discusses recurrence relations and illustrates their role in asymptotic and probabilistic analysis of algorithms. It covers in detail greedy strategies divide and conquer techniques, dynamic programming and max flow - min cut theory for designing algorithms and illustrates them using a number of well-known problems and applications.																					
8	Teaching and Learning Scheme (in Hours per Semester)	<table border="1"> <thead> <tr> <th colspan="2">Classroom Instructions (CI)</th><th>Lab Instructions (LI)</th><th>Teamwork (TW) and Self Learning (SL) (TW+SL)</th><th>Total number of Hours</th><th>Total Credits TH/30</th></tr> <tr> <th>L</th><th>T</th><th>P</th><th>SL</th><th>TH</th><th></th></tr> </thead> <tbody> <tr> <td>45</td><td>0</td><td>30</td><td>45</td><td>120</td><td>4</td></tr> </tbody> </table>				Classroom Instructions (CI)		Lab Instructions (LI)	Teamwork (TW) and Self Learning (SL) (TW+SL)	Total number of Hours	Total Credits TH/30	L	T	P	SL	TH		45	0	30	45	120	4
Classroom Instructions (CI)		Lab Instructions (LI)	Teamwork (TW) and Self Learning (SL) (TW+SL)	Total number of Hours	Total Credits TH/30																		
L	T	P	SL	TH																			
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9	Outline syllabus				CO Mapping																		

	Unit 1	Introduction	
	A	Introduction: Algorithms, Analyzing algorithms, Complexity of algorithms, Growth of functions, Performance measurements	CO1, CO3
	B	Asymptotic Notations and their properties – Mathematical analysis for Recursive and Non-recursive algorithms, Recurrences relations, Master Method	CO1, CO2, CO3
	C	Divide-and-conquer: Analysis and Structure of divide-and-conquer algorithms, Divide-and-conquer examples-Quick sort, Merge sort, Sorting in Linear Time counting sort, Heap Sort	CO1, CO2, CO4
	Unit 2	Greedy Method	
	A	Overview of the Greedy paradigm, Analysis and example: task scheduling,	CO1, CO2, CO3
	B	Fractional Knapsack problem, Single source shortest paths problem: Dijkstra's Algorithm, Bellman-ford Algorithm,	CO1, CO2, CO3
	C	Overview and analysis of Backtracking & Branch and Bound: N-Queens problem and Sum of subsets	CO1, CO2, CO3
	Unit 3	Dynamic Programming	
	A	Overview, Difference between dynamic programming and divide and conquer, All pair shortest path problems: Floyd-Warshall Algorithm	CO1, CO2, CO3, CO4
	B	Applications and analysis: Matrix Chain Multiplication,	CO1, CO2, CO3, CO4
	C	Applications and analysis: Longest Common sub-sequence, 0/1 Knapsack Problem	CO1, CO2, CO3
	Unit 4	Advanced Data Structures	
	A	Red-Black Trees - Definition, Applications, Insertion and deletion of elements in RB-Tree	CO1, CO2, CO3, CO4
	B	B-Trees - Definitions, Applications, Insertion and Deletion in B-Trees	CO1, CO2, CO4
	C	Data Structure for Disjoint Sets – Definition, Binomial Heaps, Fibonacci Heaps.	CO1, CO2, CO3, CO4
	Unit 5	Selected Topics	
	A	Introduction to NP Complete and NP Hard Problems, Examples, Amortized Analysis	CO1, CO2, CO3

	B	Approximation Algorithms – Travelling Sales Person Problem and Vertex Cover Problem, Randomized Algorithms, Randomized Quick Sort Algorithm			CO1, CO2, CO3
	C	String Matching Algorithms – Naive String Matching Algorithm, Rabin Karp Algorithm.			CO1, CO2, CO3, CO4
10	List of activities for Self Learning (SL: 45h)	S. No.	Name of the Activity	No of Hours	Evaluation Criteria
		1	Assignment writing on estimation, scheduling, earned value, and risk analysis	5 assignments of 3h each. Total = 15h	Based on the assignment submitted.
		2	Case Study Based Learning	Case Study Selection & Research =5h Analysis & Report Writing = 7h Presentation Preparation = 3 h; Total = 15h	Correct identification and use of relevant algorithms and their run times Use of standard references, textbooks, NPTEL, IEEE/Springer papers
		3	Technical Video based learning related to software project management	Duration of video = 5h; Report preparation & Presentation = 10h; Total = 15h	Report/presentation based on the video learning outcomes.
11	Mode of examination	Theory			
	Weightage Distribution	CA	MSE	ESE	
		25%	15%	60%	
12	Textbook/s*	1. Cormen <i>et al.</i> , “Introduction of Computer Algorithms”, Prentice Hall India			
	Other References	1. Sahni <i>et al.</i> , “Fundamentals of Computer Algorithms”, Galgotia Publications. 2. Hopcroft A, The Design and Analysis Computer Algorithms, Addison Wesley			

CO and PO Mapping

S. No.	Course Outcomes (COs)	Related POs/PSOs
1.	CO1: Analyze the asymptotic performance of algorithms	PO1, PO2, PO4, PO5, PO11, PSO1, PSO2
2.	CO2: Describe the dynamic-programming and Greedy paradigm and explain when an algorithmic design situation calls for it	PO1, PO2, PO3, PO5, PO11, PSO1, PSO2
3.	CO3: Demonstrate a familiarity with major algorithms and data structures	PO1, PO2, PO3, PO5, PO11, PSO1, PSO2
4.	CO4: Apply important algorithmic design paradigms and methods of analysis	PO1, PO2, PO3, PO4, PO5, PO11, PSO1, PSO2, PSO3
5.	CO5: Discuss NP-complete problems and develop algorithms to solve the problems	PO1, PO2, PO3, PO4, PO5, PO11, PSO1, PSO2, PSO3

6.	CO6: Choose appropriate algorithm design techniques for solving problems	PO1, PO2, PO3, PO4, PO5, PO9, PO11, PSO1, PSO2, PSO3
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PO and PSO mapping with level of strength for Course Name: Design and Analysis of Algorithms (Course Code: CSE356)

CSE 356	COs \ POs/PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
	CO1	3	3	1	2	2	–	–	–	–	–	2	3	2	–
	CO2	3	3	2	1	2	–	–	–	–	–	1	3	2	–
	CO3	3	2	2	1	3	–	–	–	–	–	1	3	3	–
	CO4	3	3	3	2	3	–	–	–	–	–	2	3	3	1
	CO5	3	3	2	2	2	–	–	–	–	–	2	3	2	1
	CO6	3	3	3	2	3	–	–	–	1	–	2	3	3	1

CSE356	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
	3	2.83	2.17	1.67	2.5	0	0	0	1	0	1.67	3	2.5	1

Strength of Correlation: 21.3

1. Addressed to Slight (Low=1) extent
2. Addressed to Moderate (Medium=2) extent
3. Addressed to Substantial (High=3) extent

Course-to-SDG Matrix

Sustainable Development Goals (SDGs) are 17 global targets established by the United Nations in 2015 to end poverty, protect the planet, and ensure prosperity and peace for all by 2030. The Design and Analysis of Algorithms course strongly aligns with SDG 4 by developing mathematical reasoning, analytical thinking, and advanced problem-solving skills essential for computer science education. It also strongly supports SDG 9 through the study of efficient algorithmic paradigms, optimization methods, and scalable computational techniques that contribute to innovation in modern software and computing systems. The course moderately contributes to SDG 8 by improving technical competencies and programming expertise relevant to careers in software engineering, data science, and technology industries. Additionally, it has a limited but meaningful alignment with SDG 12 because efficient algorithm design helps optimize computational resources and supports sustainable computing

practices.

Mapping uses a 3-point scale:

- Addressed to *Substantial (High=3) extent*: Core content directly contributes
- Addressed to *Moderate (Medium=2) extent*: Supports through skills/activities
- Addressed to *Slight (Low=1) extent*: Indirect awareness/benefit

SDG Goal	Description	Mapping Strength	Justification
SDG 4	Quality Education	3	Develops analytical thinking, computational problem-solving, and mathematical reasoning skills essential for advanced computer science education and research.
SDG 9	Industry, Innovation and Infrastructure	3	Supports innovation in computing systems through efficient algorithm design, optimization techniques, and scalable computational solutions used in modern technologies.
SDG 8	Decent Work and Economic Growth	2	Enhances employability by strengthening programming, analytical, and algorithmic problem-solving skills required in software development and technology-driven industries.
SDG 12	Responsible Consumption and Production	1	Efficient algorithms contribute indirectly to optimized resource utilization, reduced computational overhead, and energy-efficient software systems.

Total Mapped SDGs: 4 out of 17.